

RUSSELLVILLE 4 SSE, AL, USA

WMO: 750060

Lat: 34.4535N

Lon: 87.7104W

Elev: 720

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	20.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	92.7	N/A	90.7	N/A	88.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	DB	8.7	95.7	5.3	3.7	4.9	98.4	1.8	100.5	-1.2	102.6	-5.1	105.3	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	39.9	44.5	52.4	60.2	68.0	75.5	77.5	77.0	71.6	60.5	49.3	44.7	(6)
		DBStd	15.52	11.41	10.98	10.04	7.88	6.67	4.03	3.77	4.00	6.12	8.66	9.13	10.14	
		HDD50	1047	355	218	96	12	0	0	0	0	13	121	232		
		HDD65	3314	779	577	401	184	53	1	0	18	193	475	633		
		CDD50	4764	42	64	171	317	560	765	853	838	649	338	100	67	
		CDD65	1553	0	3	10	40	147	316	388	373	217	52	5	2	
		CDH74	14149	0	6	96	358	1289	3017	3516	3345	1954	546	21	1	
		CDH80	5398	0	0	10	40	340	1238	1459	1437	739	132	3	0	
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	60.9	5.8	5.9	5.6	5.2	5.2	5.2	4.0	3.9	3.4	5.1	6.1		
(16)		PrecMax	79.5	10.5	10.5	10.0	13.4	14.6	13.5	9.1	8.7	6.6	6.9	9.5	16.9	
(17)		PrecMin	36.6	2.2	2.6	0.8	1.5	0.7	1.4	1.8	0.2	0.6	0.0	3.1	2.5	
(18)		PrecStd	10.4	2.3	2.5	2.2	2.8	3.8	2.8	2.2	2.1	1.6	2.0	1.7	3.4	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	69.3	74.4	81.5	83.7	88.1	94.6	94.3	97.9	93.3	87.2	76.7	72.4		(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(21)		DB	65.7	70.3	76.7	81.0	86.1	91.0	91.4	93.0	90.6	83.8	71.9	68.6		
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(23)		DB	62.0	66.7	72.8	78.4	84.0	89.1	89.6	90.3	87.7	80.1	68.9	65.5		
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(25)		DB	58.6	62.6	69.0	75.4	81.7	87.0	87.7	87.7	84.6	76.6	65.8	61.7		
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(29)		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(30)		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(31)		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(32)		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(33)		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(34)		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(35)	Mean Daily Temperature Range	MDBR	19.7	20.6	22.5	23.7	21.7	21.5	19.3	20.3	22.7	24.4	24.0	18.9		(35)
(36)		5% DB	19.8	23.7	25.4	25.3	24.3	24.2	21.4	23.7	25.3	27.6	25.7	19.5		
(37)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(38)		5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(39)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.321	0.359	0.391	0.430	0.458	0.500	0.510	0.421	0.366	0.334	0.314		(40)
(41)		taud	2.511	2.490	2.409	2.309	2.243	2.187	2.100	2.062	2.303	2.440	2.472	2.529		
(42)		Ebn at Noon	284	294	290	284	273	264	252	247	266	274	274	278		
(43)		Edh at Noon	26	30	35	41	44	47	51	52	39	31	27	25		
(44)	All-Sky Solar Radiation	RadAvg	790	953	1318	1696	1872	1990	1903	1761	1535	1230	899	674		(44)
(45)		RadStd	63	121	121	115	139	156	109	103	134	161	93	67		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (30 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+169

Nomenclature: See separate page